

Journal Supplementary C — Per-axis motivation and alternatives

Companion to: *Beyond Accuracy* (Section III)

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A.1 Scope

This supplement documents, per axis F-1 through F-12, the question the axis answers, the metric chosen, the alternatives considered, and the design rationale for the final selection. The intent is that a reader could substitute an alternative metric within an axis without re-architecting the framework.

B.2 F-1: Hierarchical Bayesian classification

Question. “Does the topic representation outperform non-topic baselines after pooling across scenes and folds?”

Metric. Posterior mean and HDI94 of μ_m in the hierarchical model $\text{score}_{m,s,f} = \mu_m + \delta_s + \rho_f + \epsilon$, plus the pairwise $P[\mu_A > \mu_B]$ table.

Alternatives considered. (a) Pooled OA average (drops scene-level uncertainty); (b) Fixed-effect ANOVA (would require asymptotic normality on small per-fold samples); (c) Bayesian paired-tests per (scene, method) (high false-discovery rate without correction).

Rationale. The hierarchical model is the smallest design that (i) returns a per-method posterior, (ii) marginalises over scene-level offsets, (iii) yields pairwise probabilities directly interpretable as a graded ordering. The NUTS sampler with 1000 draws + 1000 tune + 2 chains is the default PyMC configuration; no divergences are observed.

C.3 F-2: Topic-word coherence

Question. “Are the top- N words per topic semantically coherent under the corpus’s own co-occurrence statistics?”

Metric. c_v [1], c_{NPMI} , and U-Mass, reported with $N = 15$ and a sliding window of size 110 (matching Röder’s default).

Alternatives considered. (a) UCI coherence (less correlated with human judgement on Röder’s benchmarks); (b) c_p Pointwise mutual information (drops the sliding window); (c) Word2Vec embeddings (requires an external corpus, breaks the self-contained reproducibility commitment).

Rationale. c_v is the strongest performer in Röder 2015 on human judgements; reporting NPMI alongside is a robustness check; U-Mass is the historical reference. The three-way report also enables the coherence-vs-ARI scatter that surfaces the LDA-vs-ProdLDA tradeoff.

D.4 F-3: Seed stability

Question. “Does refitting under a different random seed yield the same topics?”

Metric. Mean $\pm$ std ARI(KMeans-on- θ , label) over $N = 5$ seeds plus mean $\pm$ std c_v over the same five.

Alternatives considered. (a) Topic-by-topic matched cosine (sensitive to permutation, requires pre-alignment); (b) Bootstrap-resampled corpora (mixes seed stability with sample stability); (c) $N = 30$ seeds (the companion code repository’s `deep_seed_stability/` block runs $N = 30$; the framework uses $N = 5$ for parsimony plus an explicit upgrade path).

Rationale. The ARI + c_v pair is the smallest set that surfaces both the cluster-quality and topic-interpretability sides of stability. The five-seed budget is enough to estimate the std reliably for a unimodal posterior; multi-modal posteriors (which we suspect on KSC and Botswana) would benefit from larger N and are flagged for follow-up.

E.5 F-4: Capacity sensitivity

Question. “How does the topic representation change when one allows K to vary?”

Metric. ARI(canonical, KMeans-on- θ_K , label), held-out perplexity, topic diversity (fraction of unique top-15 words), and cross-seed matched-cosine, all reported across $K \in \{4, 6, 8, 10, 12, 16, 20, 24\}$.

Alternatives considered. (a) ELBO vs K curve (diverges in batch sizes and not directly interpretable); (b) AIC / BIC at the corpus level (penalty terms become arbitrary for non-Gaussian likelihoods); (c) Held-out perplexity alone (the de-facto default but uninformative on topic interpretability).

Rationale. The four-metric report covers the three distinct ways capacity choice can fail: underfitting (perplexity high, ARI low), overfitting on tokenisation noise (perplexity low, diversity collapses), and topic splitting/merging without ARI gain (matched-cosine high but ARI flat).

F.6 F-5: Band-mask robustness

Question. “Does the topic basis survive a class of spectral-window restrictions?”

Metric. Paired ARI between canonical and masked dominant-topic maps, restricted to pixels labelled in both, under four canonical masks (VNIR, SWIR, no-water, top-50 Fisher).

Alternatives considered. (a) Phi-cosine after Hungarian alignment (basis-level, but conflates marginal-prevalence with identity); (b) KL between θ posteriors per pixel (continuous, but interpretation across pixels is unclear); (c) Jaccard of top-words per topic after Hungarian alignment (vocabulary-level, useful supplementary B axis in the conference paper).

Rationale. Paired ARI is the metric most directly interpretable as “how much agreement under best-permutation” on the same data set; it is the principal metric. The companion swap-rate is the un-corrected fraction and the Hungarian alignment is the per-topic correspondence; the three together constitute the axis output.

G.7 F-6: Cross-method clustering agreement

Question. “How does the canonical topic-dominant map relate to other partitions of the same scene?”

Metric. All-pairs ARI and NMI matrix between {label, topic-dominant, Felzenszwalb, SLIC-500, SLIC-2000, patch-15, patch-7, pixel}.

Alternatives considered. (a) Pair-by-pair ARI with a sole reference (loses the cross-method symmetric structure); (b) V-measure heatmap (V-measure equals NMI on these partitions and gives the same information); (c) Graph-clustering modularity between methods (interesting but requires a per-method graph that we do not produce).

Rationale. The 8×8 matrix surfaces shared inductive biases (e.g. SLIC-500 and patch-15 are spatially smooth and agree highly even though they disagree with the ground-truth label) that a single-row reference cannot show.

H.8 F-7: Topic-label coupling

Question. “How concentrated is the support of each canonical topic on the ground-truth label?”

Metric. Per-topic $H(L | t) = -\sum_{\ell} P(\ell | t) \log P(\ell | t)$ and the corresponding KL divergence $\text{KL}(P(L | t) \| P(L))$.

Alternatives considered. (a) Topic-to-label hard mapping via Hungarian (a special case of F-8 applied to the label side); (b) Mutual information $I(L; T)$ (a corpus-level summary, useful as a single-number companion); (c) Topic precision/recall on the dominant-label hypothesis (per-(topic, label) but multiplies the artefact size).

Rationale. Per-topic H and KL admit direct interpretation as concentration / divergence from the marginal, and integrate cleanly with the dominant-topic-map raster that the framework already produces.

I.9 F-8: Per-topic identity tracking under masks

Question. “Which masked topic corresponds to which canonical topic, and how much disagreement survives the correspondence?”

Metric. Hungarian permutation $\sigma^* = \arg \max_{\sigma} \sum_k C_{k, \sigma(k)}$ on the confusion matrix $C \in \mathbb{Z}_{\geq 0}^{K \times K^{(m)}}$, plus the swap rate under alignment $|\mathcal{S}|^{-1} \sum_d \mathbf{1}[z_d^{(m)} \neq \sigma^*(z_d)]$.

Alternatives considered. (a) Top- N greedy assignment (faster but sub-optimal); (b) Soft alignment via Sinkhorn (returns a transport matrix, not a permutation); (c) Spectral alignment of the eigenvectors of C (numerical stability concerns at small K).

Rationale. The exact Hungarian solution is computable in $O(K^3)$ on $K \leq 12$ in milliseconds and is the standard benchmark in the cluster-comparison literature.

J.10 F-9: Preprocessing stability (HIDSAG)

Question. “Does the topic basis survive a change in the preprocessing pipeline (continuum removal, smoothing, derivative)?”

Metric. ARI(dominant-topic recipe A, dominant-topic recipe B) per pair, summarised as the mean ARI across the $\binom{R}{2} = 3$ pairs for $R = 3$ recipes.

Alternatives considered. (a) Phi-cosine pairwise (basis-level, parallel to F-5’s alternative); (b) Recipe-conditioned classification accuracy (drops the topic-level question); (c) Procrustes alignment of ϕ matrices (returns a continuous similarity but does not produce a discrete per-pair scalar).

Rationale. Mean pairwise ARI is the simplest summary that yields a per-subset scalar comparable across HIDSAG subsets.

K.11 F-10: Cross-scene topic transfer

Question. “Does the canonical topic basis on scene s generate a useful topic representation on scene t ?”

Metric. ARI(label $_t$, dominant-topic of $\theta^{(t)} = \theta(\phi^{(s)}, \mathbf{c}^{(t)})$) for the transfer $\phi^{(s)} \rightarrow \theta^{(t)}$.

Alternatives considered. (a) Held-out perplexity of $\phi^{(s)}$ on scene t (loses the cluster-quality side); (b) Top-words drift between $\phi^{(s)}$ and a refit $\phi^{(t)}$ (a basis-level statistic that does not say whether the transfer is useful downstream).

Rationale. The ARI report is the most directly interpretable transfer score and is symmetric in the sense of producing both $s \rightarrow t$ and $t \rightarrow s$ numbers.

L.12 F-11: Rate–distortion of θ

Question. “How aggressively does the topic representation compress the spectrum while remaining useful?”

Metric. Per-pixel MSE($\hat{\mathbf{x}} = \phi^\top \theta, \mathbf{x}$) versus rate $\log_2 K$.

Caveat. The reconstruction $\hat{\mathbf{x}}$ uses the quantised band-frequency tokenisation; the MSE therefore mixes a tokenisation-loss component into the rate–distortion curve. We flag this as a limitation in journal §VII and as a candidate for follow-up axis design.

M.13 F-12: External baseline comparison

Question. “How does the canonical topic-routed-soft OA compare to the literature on the same scene?”

Metric. OA delta = topic-routed-soft OA – best published OA per scene, where “best published” is taken from a fixed set of three reference papers per scene.

Caveat. Train/test splits across papers are not always comparable, so the delta is reported with the per-paper split caveat in journal §VII and in the artefact.

References

- [1] M. Röder, A. Both, and A. Hinneburg, “Exploring the space of topic coherence measures,” in *Proceedings of the Eighth ACM International Conference on Web Search and Data Mining (WSDM)*, 2015, pp. 399–408.